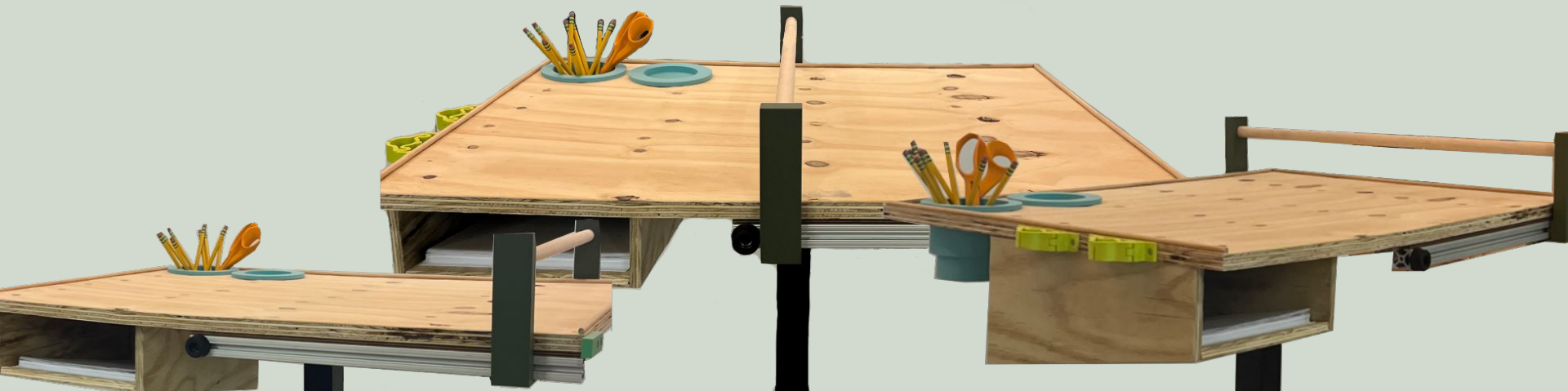
# Mr. Collins' Desk

ME40 User-Centered Design Project

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Spring 2025



# Challenge Overview

Mr. Collins is an English and Social Studies teacher at Somerville Middle School. He has cerebral palsy and uses a standing desk as a mobility aid in the hallways at school. The desk does not have adequate storage for items he needs to bring with him, such as papers and writing materials. Additionally, it is inconvenient to place his walking sticks in their current holders as they stick out from the desk and limit ease of maneuverability of the desk. Lastly, it is difficult for him to lower the desk by himself, and keep the desk stable when it has items on top.



# Existing Product Research

















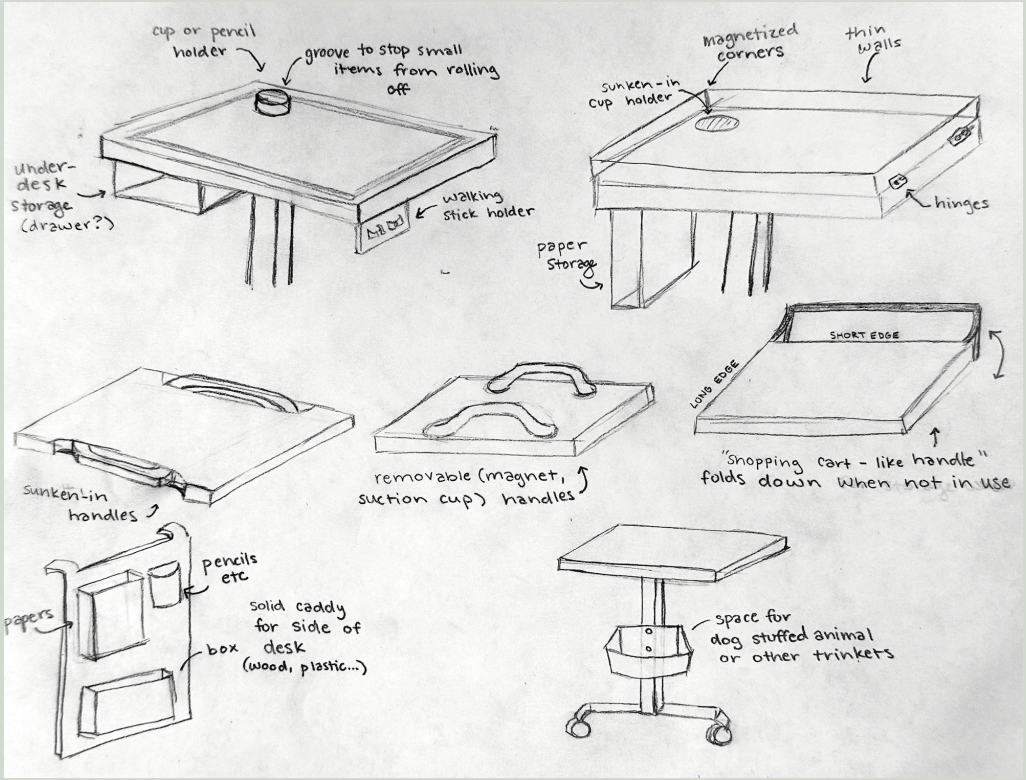

# User Needs and Engineering Requirements

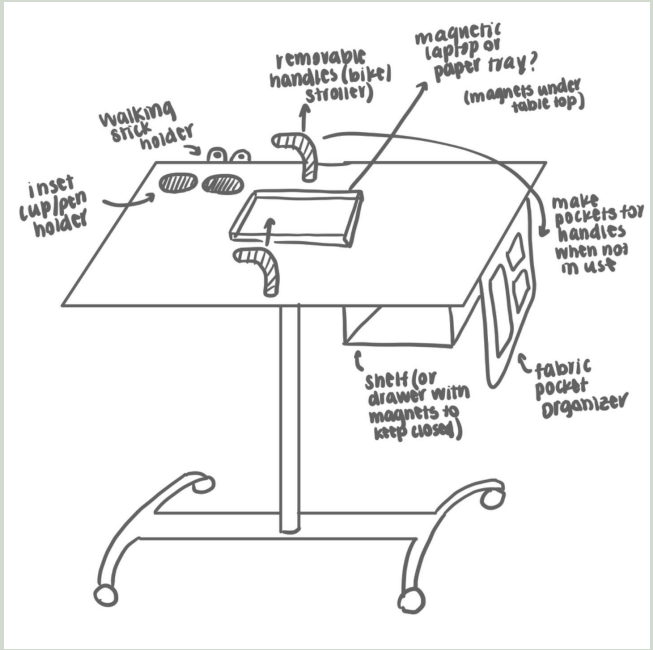
| User Need | Engineering Requirement |
|---|---|
| **1. Stability-** lean on while walking | - Maximum weight capacity: 150 lbs, direction: ^<br>- No attachments obstruct ability to lean body weight on desk via forearms |
| **2. Storage**- papers, walking sticks, etc | - Storage space: 8.5x11x2 (paper space), direction:^<br>- Incorporation of an easily accessible (one step) walking stick holder which does not impede the movement of Mr. Collins with the desk |
| **3. Convenience**- mechanism that prevents things from falling off | - Height- prevent taller objects (stationary holder, water bottle, etc) from falling off |
| **4. Maneuverability**- get around/over hallway obstacles | - Less than two obstacles (eg gaps or floor irregularities) slow Mr. Collins down when walking with the desk per minute of normal school travel |
| **5. Attachable**- things we design should add on to desk | - Add-ons are securely attached, all can withstand minimum 5 lbs of force tangent to attachment surface |

# User Needs and Engineering Requirements II

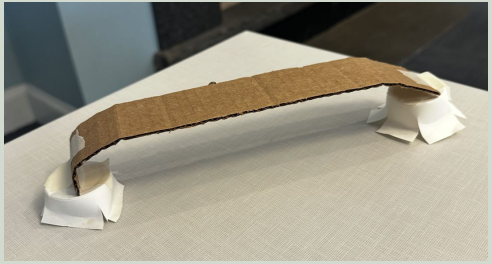
| User Need | Engineering Requirement |
|---|---|
| **6. Adjustable**- to correct heights | - Desk can be adjusted to any desired height by Mr. Collins with no help from additional people |
| **7. Fun**- for both user and students, incorporated student ideas and engages them in design process | - Design incorporates at least one element aimed at the students: decorative element or student design idea |
| **8. Ease of use**- easy to raise and lower | - Able to raise and lower while sitting<br>- See #6, “adjustable” |
| **9. Versatile/interchangeable**- serve different purposes as needed, anything we add can be edited or changed | - No adjustments permanently modify the desk<br>- Mr. Collins (with up to one helper) can interchange what add-ons are attached to the desk |
| **10. Aesthetic**- non-distracting design for classroom | - When polled, people rate it as “classroom suitable” and “good looking” |

# Ideation

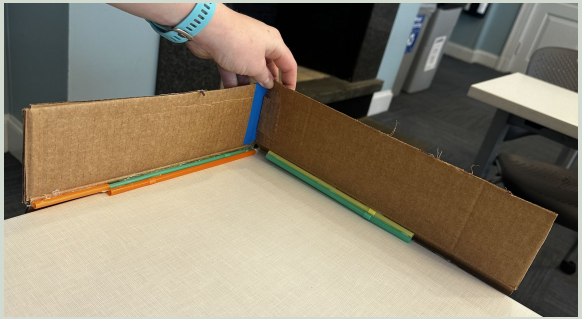

cup or pencil holder
groove to stop small items from rolling off
under-desk storage (drawer?)
walking stick holder
magnetized corners
thin walls
sunken-in cup holder
hinges
paper storage
sunken-in handles
removable (magnet, suction cup) handles
SHORT EDGE
LONG EDGE
"shopping cart-like handle" folds down when not in use
pencils etc
papers
box
solid caddy for side of desk (wood, plastic...)
space for dog stuffed animal or other trinkets


removable handles (bike/stroller)
magnetic laptop or paper tray?
(magnets under table top)
walking stick holder
inset cup/pen holder
make pockets for handles when not in use
shelf (or drawer with magnets to keep closed)
fabric pocket organizer

# Low Fidelity Prototypes



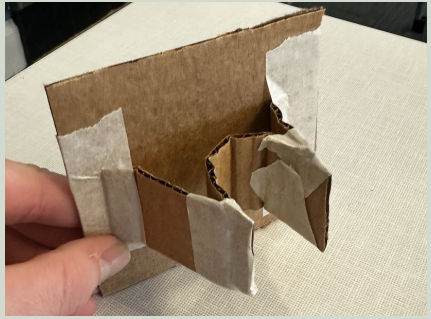
Suction cup handle

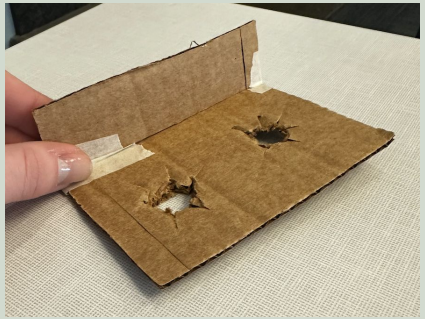


Desk edge walls



Under-desk shelf
with door



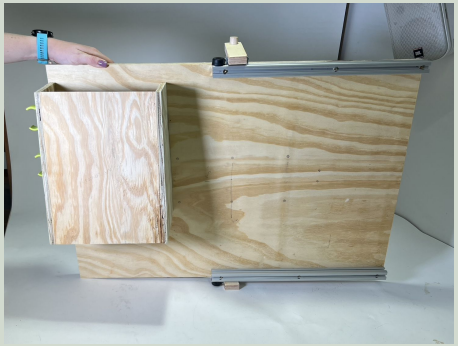


Walking stick holders

# Medium Fidelity Prototype











# Final Prototype









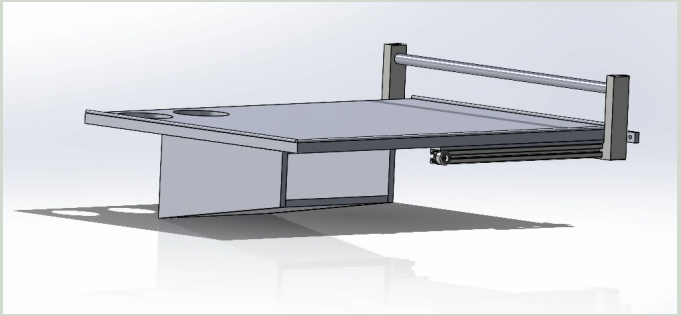


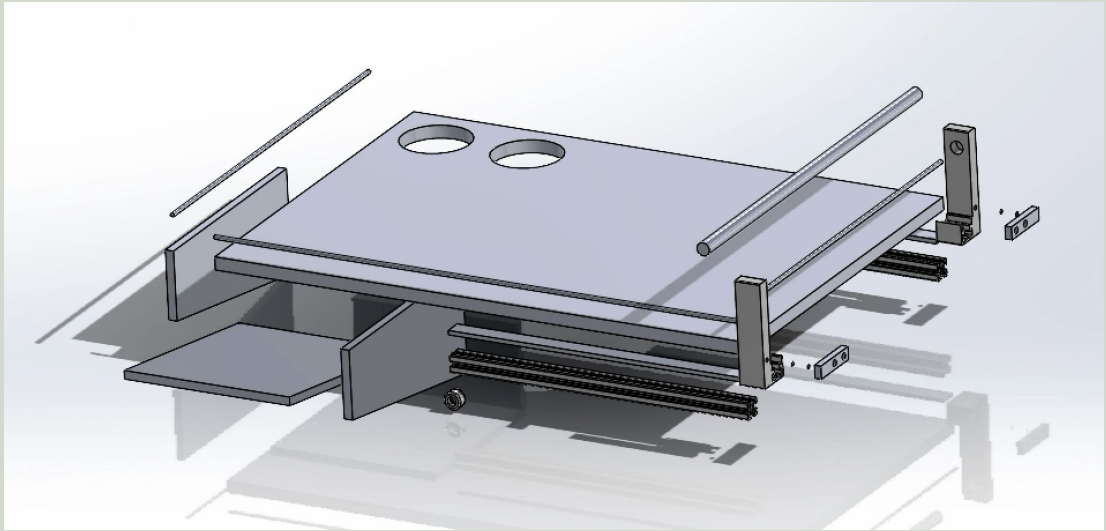


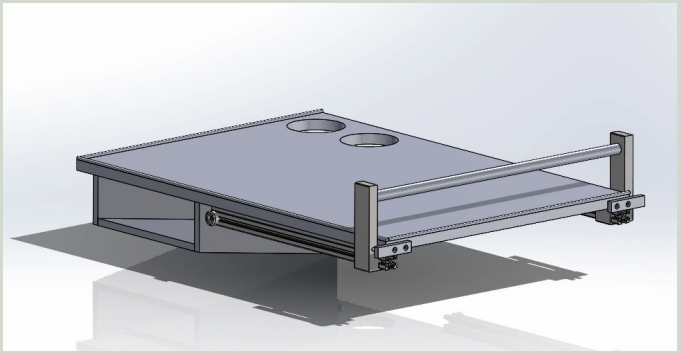


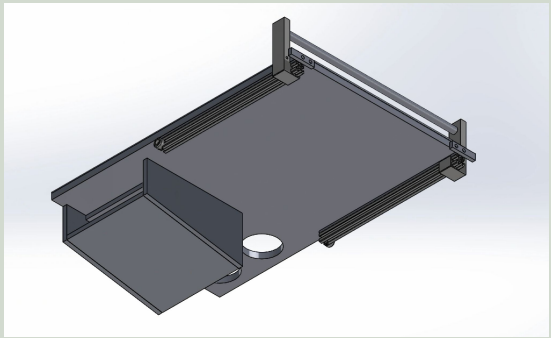


# Final Prototype









# Final Prototype Features

- Sliding **handle** for stability while walking
  - Mounted on 1 inch 8020 channels with 3D printed mounts
  - Bumpers keep the handle engaged and magnets keep handle to the side when not in use
- 3D printed **walking stick holders** keep them accessible but out of the way
- 10 degree slanted wooden **shelf** holds a full stack of papers
  - No hinges or doors for easy access to paper
- Inset mug and **cup holders** for additional storage
- ¼” dowel **ridges** to prevent items from sliding off the desk when in motion
- 3D printed **chair** for stuffed guide dog Bonnie
- **Heavy plywood desktop** for easy lowering

# Future Directions

While this we are really excited about our designs and features of our prototype, there is still a lot we would love to improve and have ideated throughout the design process so far.

One of the main things we would love to improve is the ease of use of the sliding handle. Right now, there is a lot of friction between the handle and the 80-20 metal, which makes it difficult to slide the handle back and forth. For future designs with more budget, we would like to use metal linear bearings rather than the 3D printed ones we have right now, in hopes that the slippery plastic lining would decrease friction.





# Future Directions II

We would also like to have a better finish on the desk and it’s components for a final product. This would include a wood varnish over the entire thing and higher quality wood than plywood (such as birch or oak) to have a nicer looking desk. Unfortunately budget and supply limitations prevented us from being able to achieve this for the prototypes.

# Video

